

Work Experience

Cesium, 3D Graphics Engineer

June 2025 - current

- Build core rendering systems for CesiumJS, a WebGL engine for massive 3D geospatial datasets.
- Optimize performance across terrain traversal, picking, billboard rendering and other workflows.
- Design and develop custom modules for 3D mesh editing capabilities on top of CesiumJS.

University of Pennsylvania, GPU Programming Lecturer

August 2026 - current

- Teach CUDA and modern graphics APIs to 60 students through twice-weekly lectures and labs.
- Manage and mentor four teaching assistants, overseeing recitations and office hours.
- Maintain and improve course material and organize guest lectures with industry experts.

Amazon Web Services, Software Development Engineer

April 2022 - August 2024

- Developed Tying, a Java-based engine for automated patching and OS migration of Amazon hosts.
- Designed and authored integration and end-to-end load tests for the Tying engine.
- Improved DevOps tooling and monitored service metrics proactively and in on-call capacity.
- Expanded and maintained CI/CD and distributed cloud computing infrastructure as code.

Epic Systems, Software Developer

August 2019 - August 2021

- Developed and designed the App Orchard website, a marketplace for third-party healthcare apps.
- Implemented RESTful web APIs to expose health record data according to the HL7 FHIR standard.
- Led transition of App Orchard site from ASP.Net framework to single-page React app.

Education

University of Pennsylvania, 2025 M.S. in Computer Graphics and Game Technology, 3.93 GPA

Rice University, 2019 B.S. in Computational Physics; Minor in Applied Mathematics, 3.88 GPA

Projects

Cubit (Maya Plugin), <https://github.com/mzschwartz5/cubit>

- Soft-body deformation and destruction in a single plugin using PBD with voxel gram-schmidt constraints, and custom tools for painting simulation mesh constraints and live interaction.

Fluid Surface Mesh Shader, <https://github.com/mzschwartz5/Breakpoint>

- A mesh-shading based fluid surface reconstruction demonstration with a 4x performance improvement compared to the original (<https://dl.acm.org/doi/abs/10.1145/3651285>).

Node Based SDF Modeling Tool, <https://github.com/mzschwartz5/Node-Based-SDF-Modeling-Tool>

- An OpenGL node-graph tool for quickly creating procedural 3D models, via raymarched SDFs.

CUDA Pathtracer, <https://github.com/mzschwartz5/CUDA-Path-Tracer>

- A pathtracer built on CUDA with support for various PBR materials, GLTF model loading, denoising, and various performance optimizations.

Technologies: C++, DirectX, CUDA, HLSL, OpenGL, QT, Unreal Engine, WebGPU, Unity, Maya